

Multi-Corpora Digital Twin Framework

1. Architectural Definition

A multi-corpora ecosystem bridges four distinct technological languages into a unified operational digital twin:

- **Code Corpus:** Syntax trees, interfaces, native Kotlin/Java Android services, Python orchestration scripts.
- **DevOps/Infrastructure Corpus:** Shell initialization scripts, Termux environment configs, Tailscale mesh routes, ADB bridges.
- **Asset Corpus:** Model weights (Qwen, Gemma GGUF), token matrices, audio ONNX models, UI vector assets.
- **Human Knowledge Corpus:** Conceptual notes, system specifications, living wiki graphs, task ledgers.

2. Federation Mechanisms

1. **Federated Identity:** Global metadata ontology linking abstract entities across systems using deterministic UUIDs.
2. **Semantic Cross-Compilation:** Translating code tokens and non-text asset descriptors into shared vector embedding spaces for universal multimodal retrieval.
3. **Reactive Change Propagation:** Event-driven synchronization ensuring modifications in code automatically trigger re-evaluations across documentation and RAG indexes.